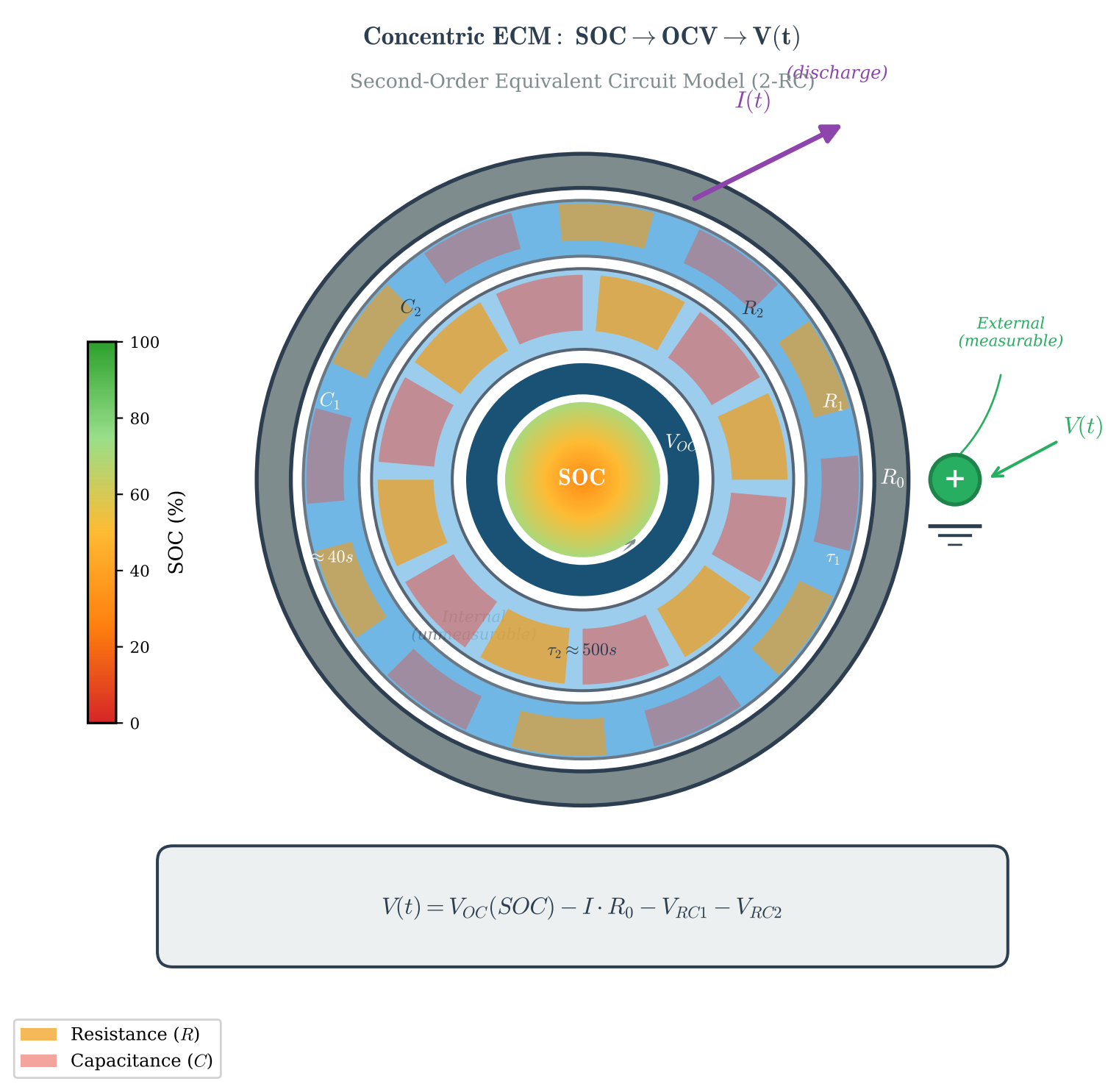

Concentric ECM : SOC → OCV → V(t)
Second-Order Equivalent Circuit Model (2-RC)
I(t) (discharge)
External (measurable)
V(t)
SOC (%)
100
80
60
40
20
0
SOC
V_{OC}
C_2
R_2
C_1
R_1
R_0
τ_1
$\approx 40s$
Internal (unmeasurable)
$\tau_2 \approx 500s$
$V(t) = V_{OC}(SOC) - I \cdot R_0 - V_{RC1} - V_{RC2}$
Resistance (R)
Capacitance (C)